



FIELD UNIT • POLLINATOR & BIODIVERSITY MONITORING

BeeMonitor

HARDWARE ASSEMBLY MANUAL

A tripod-mounted camera system for monitoring pollinator behavior and biodiversity in the field.

CONTENTS

Parts & accessories	02
Sourcing & cost	03
Assembly · steps 1–9	04
Powering the unit · step 10	09
Next steps & support	10



PARTS & ACCESSORIES

14 PARTS • FIG. 1

Check that every part is in your kit before you start. Numbers match the labels in the photo.
Dashed parts are optional and only needed to run the unit from a battery or solar panel.



FIG. 1 • KIT CONTENTS

1 Tripod Purchased	2 Case base 3D printed	3 Case lid 3D printed	4 Tripod connector 3D printed	5 Power connector 3D printed
6 Case cable ties Purchased	7 Power button With Pi kit	8 Camera Purchased	9 Raspberry Pi 4 Purchased	10 microSD card Purchased
11 Power cable With Pi kit	12 Battery Optional	13 Solar cable Optional	14 Solar panel Optional	Required Optional (field power)

SOURCING & COST

TABLE 1 • USD

Where to get each part. Prices are in US dollars at the time of writing and will vary by supplier and region. Shaded rows are optional field-power parts.

NO.	PART	STATUS	SOURCE	PRICE
01	Tripod	Required	amazon.com/dp/B00XI87KV8	20.00
02	Case base	Required	3D printed	2.50
03	Case lid	Required	3D printed	1.50
04	Tripod connector	Required	3D printed	0.50
05	Power connector	Required	3D printed	0.50
06	Case cable ties	Required	amazon.com/dp/B0DNF7WDSW	2.00
07	Power button	Required	Included with Pi kit	incl.
08	Camera	Required	amazon.com/dp/B0D3XH29S8	57.00
09	Raspberry Pi 4	Required	amazon.com/dp/B07TXKY4Z9	125.00
10	microSD card	Required	amazon.com/dp/B0B7NV73PJ	48.00
11	Power cable	Required	Included with Pi kit	incl.
12	Battery	Optional	amazon.com/dp/B0CQT5G1ZR	80.00
13	Solar cable	Optional	amazon.com/dp/B0GLH13JK5	56.00
14	Solar panel	Optional	Included with solar kit	incl.

CORE BUILD • REQUIRED PARTS

\$257.00

Mains-powered unit, parts 1–11

FULL FIELD BUILD • ALL PARTS

\$393.00

Adds battery and solar power, parts 12–14

3D-printable case files: github.com/eai6/BeeMonitor/tree/main/hardware/enclosure

PROGRESS

1

2

3

4

5

6

7

8

9

10

STEP 1 OF 10

1

STEP 01 • USES PARTS 8, 9

CONNECT THE CAMERA

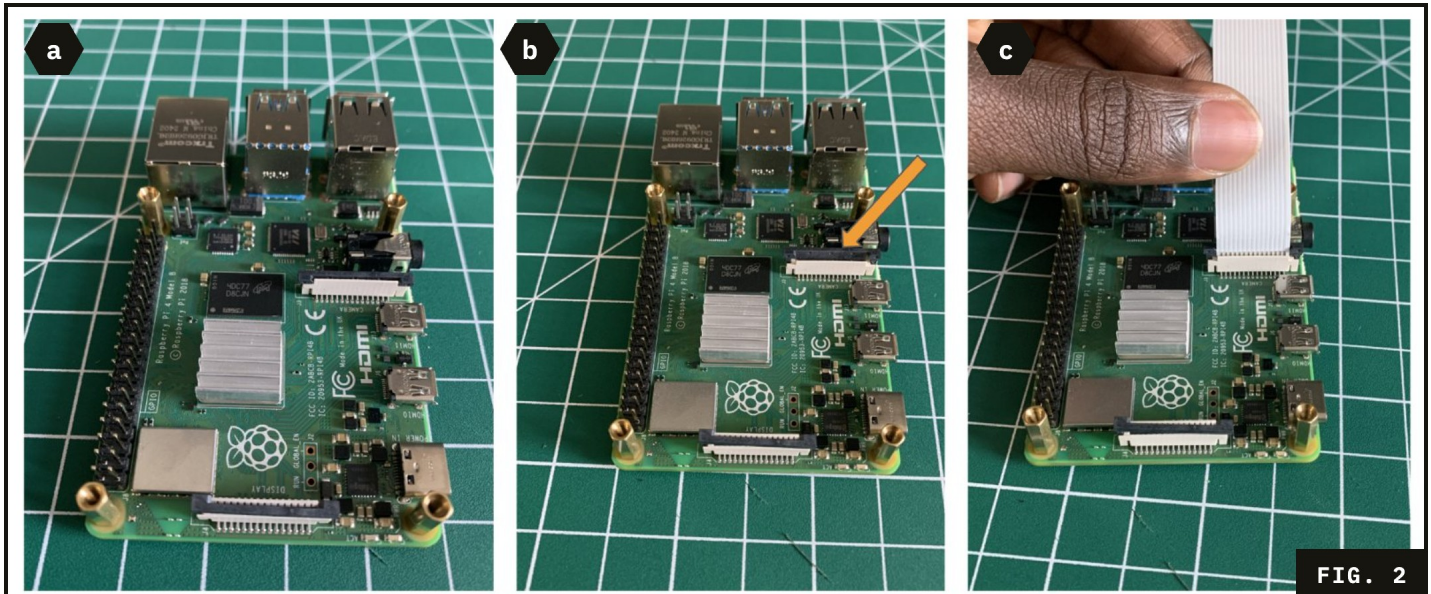


FIG. 2

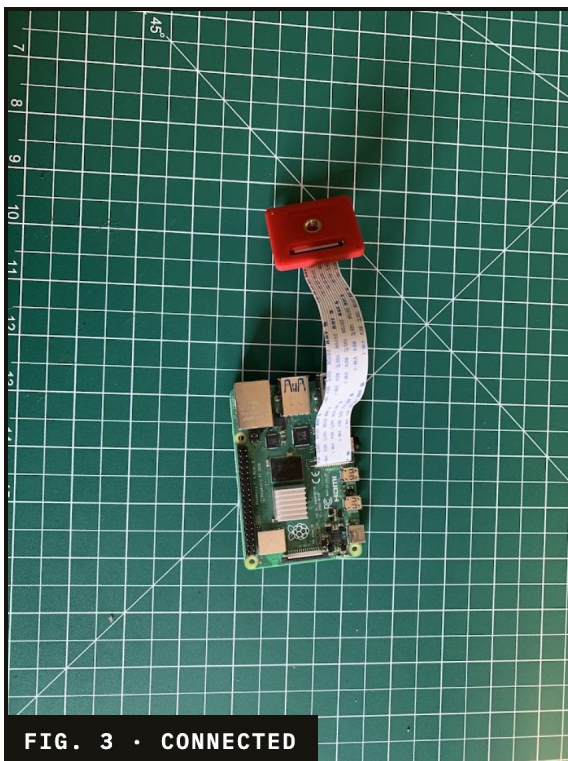


FIG. 3 • CONNECTED

Insert the camera ribbon into the camera slot of the Raspberry Pi 4.

- a** Lay the Raspberry Pi 4 face up.
- b** Find the camera slot (arrowed) and gently lift its plastic latch.
- c** Push the ribbon fully into the slot, then press the latch down to lock it.

i **Tip** • Hold the ribbon by its edges and avoid sharp folds.

PROGRESS

1

2

3

4

5

6

7

8

9

10

STEPS 2-3 OF 10

2

STEP 02 · PARTS 2, 8, 9

SEAT IN THE CASE BASE

Slide the camera and Raspberry Pi into the case base.



FIG. 4



Tip · Keep the ribbon flat so it is not pinched when the lid goes on.

3

STEP 03 · PART 10

INSERT THE SD CARD

Insert a BeeMonitor-flashed and activated microSD card into the Raspberry Pi 4.



FIG. 5



Before this step · Flash and activate the card by following the Add Device page: beemonitor.edwardamoah.com/devices/enrollment

PROGRESS

1

2

3

4

5

6

7

8

9

10

STEPS 4-5 OF 10

4

STEP 04 · PART 7

FIT THE POWER BUTTON

Insert the power button through the side of the case.



Tip · Fit the Pi into the case with the cable side facing up.

5

STEP 05 · PART 3

CLOSE THE CASE

Cover the case base with the lid.



Tip · Check that no cable is caught under the edge of the lid.

PROGRESS

1

2

3

4

5

6

7

8

9

10

STEPS 6-7 OF 10

STEP 06 • PART 4

6

ADD THE TRIPOD CONNECTOR

Screw the tripod connector into the base of the case.



FIG. 8

i **Tip** • Hand-tighten only. Printed threads can strip if forced.

7

STEP 07 • PART 6

SECURE THE LID

Fasten the lid to the case base with the cable ties.

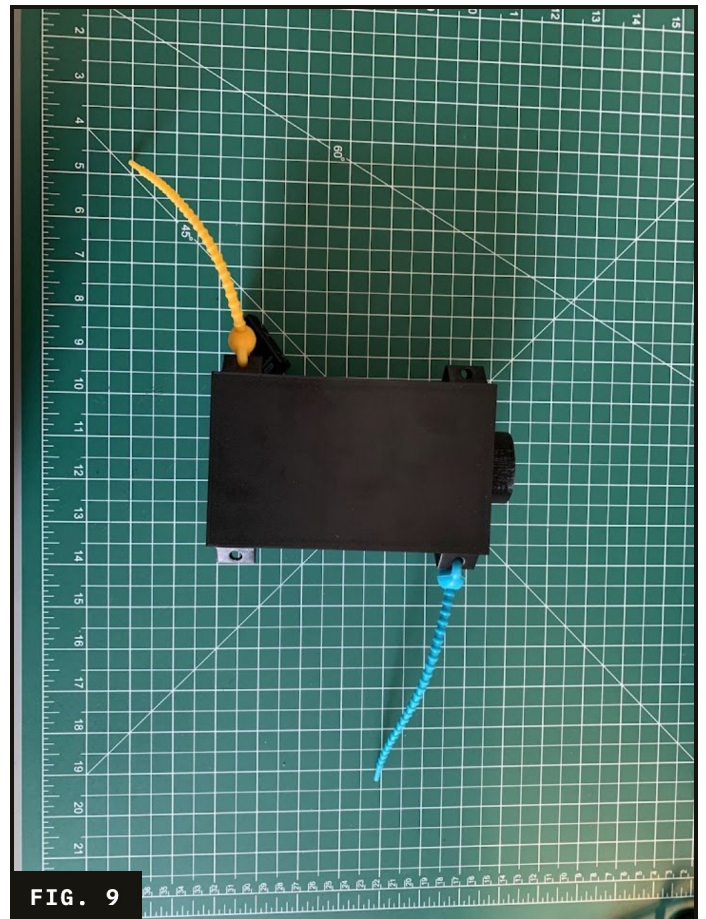


FIG. 9

i **Tip** • The silicone ties are reusable, so the case can be reopened later.

PROGRESS

1

2

3

4

5

6

7

8

9

10

STEPS 8-9 OF 10

8

STEP 08 • PARTS 1, 4

MOUNT ON THE TRIPOD

Attach the case to the tripod head using the tripod connector at the base of the case.



FIG. 10

Tip • Lock the tripod legs before letting go of the case.

9

STEP 09 • PART 11

CONNECT THE POWER CABLE

Plug the power cable into the BeeMonitor.



FIG. 11

Tip • Leave some slack so the camera can be re-aimed without pulling the cable.

PROGRESS

1

2

3

4

5

6

7

8

9

10

STEP 10 OF 10

10

STEP 10 • CHOOSE ONE POWER OPTION

POWER THE BEE MONITOR

Run the unit from a wall socket, or from a battery in the field. Add the solar panel to keep the battery charged.

A WALL SOCKET

PART 11

Plug the power cable into a power socket. Best for lab use or a site with mains power.



FIG. 12

B BATTERY

PARTS 11, 12 • OPTIONAL

Connect the power cable to the battery instead of a socket. Use this in the field.

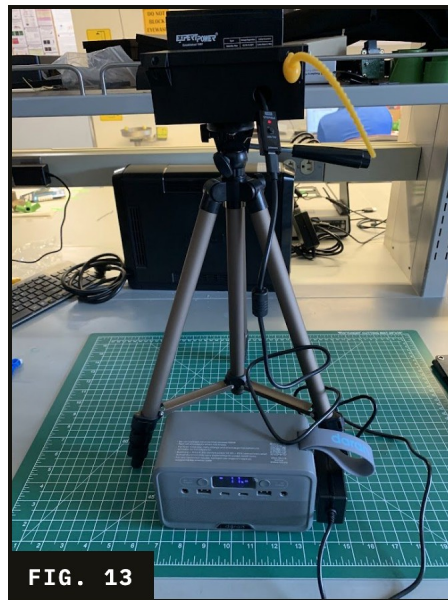


FIG. 13

C BATTERY + SOLAR

PARTS 12-14 • OPTIONAL

Set up option B, then connect the solar panel to the battery with the solar cable.

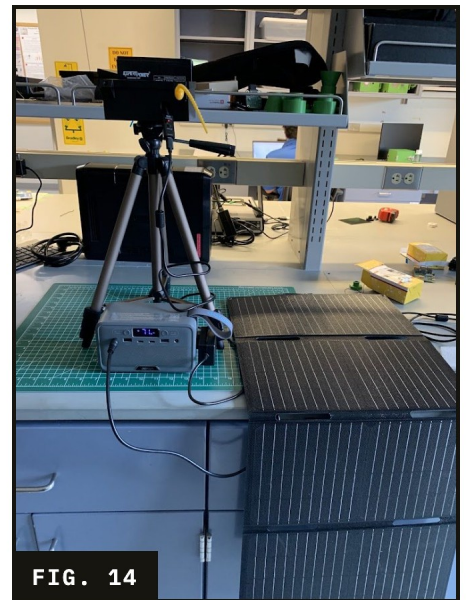


FIG. 14



Switch on • With power connected, press the power button on the side of the case to start the BeeMonitor.



Assembly complete.

Point the camera at your study site, whether a bee hotel, a flower patch or a nesting area, then start recording from the BeeMonitor platform.

PLATFORM

Add devices, flash SD cards and view recordings.

beemonitor.edwardamoah.com

PRINT FILES

Case base, lid and connectors for 3D printing.

github.com/eai6/BeeMonitor
or/ [hardware/enclosure](#)

SUPPORT

Missing a part or stuck on a step? Get in touch.

eai6@psu.edu

FINISH